# SpendWise

## Lean Canvas and Concept Note

Personal Expense Analysis Dashboard | Undergraduate Web Development Project

This document describes the proposed value, scope, methodology, and limitations of SpendWise. It distinguishes project assumptions from outcomes that have not yet been validated.

### Project at a glance

**Purpose:** Help users manually record personal expenses, review category patterns, and monitor a monthly budget in a simple browser-based interface.

**Technology:** HTML5, CSS3, vanilla JavaScript, browser local storage, and optional GitHub Pages hosting.

**Current scope:** A single-user, static prototype. No bank connection, server, account system, or cross-device synchronisation is included.

**Data transparency:** Starter transactions are fictional demonstration data. The application is not financial advice and does not verify a user's full financial situation.

# Part I — Lean Canvas

The canvas below is a planning hypothesis for the project, not evidence of validated demand or a proven business model.

### Problem

- People may forget or inconsistently record everyday spending.
- It can be hard to see which categories account for recorded expenses.
- Monthly budget use may be difficult to review at a glance.

### Existing alternatives

- Paper notes or diaries.
- Spreadsheet templates.
- General-purpose expense apps and banking apps.

**Assumption:** users may value a simple, lightweight option.

### Solution

- Manually add and categorise expenses.
- Monthly totals, daily average, and transaction count.
- Category breakdown, daily chart, budget progress, and CSV export.

### Unique value proposition

A simple, approachable expense dashboard that demonstrates the essentials without requiring an account, bank connection, or installation.

### Customer segments

Potential users: students and individuals who want a basic way to track spending. These segments have not yet been validated through interviews or surveys.

### Key metrics

- Successful expense additions and deletions in testing.
- Correctness of summary calculations.
- CSV export success rate.
- Task completion and usability feedback in a future user test.

### Channels

- College project demonstration.
- GitHub Pages project website.
- Peer or classroom feedback if collected with appropriate consent.

### Cost structure

Initial prototype: no required hosting fee on GitHub Pages for a public repository, no paid API, and no database. Future backend, security, or support services may add costs.

### Revenue or value model

Current model: educational and demonstration value, not revenue. Possible future options such as optional premium features would require user research and a separate feasibility review.

### Potential unfair advantage

No defensible advantage has been established. A focused student-friendly experience and transparent local-first design are potential differentiators, but would need comparison and validation.

# Part II — Concept Note

## 1. Project title

SpendWise: Personal Expense Analysis Dashboard.

## 2. Background and rationale

Managing day-to-day spending can be difficult when purchases are recorded inconsistently or not reviewed regularly. A lightweight expense tracker can help users organise entries and view their recorded spending in one place. A browser-based prototype is appropriate for an undergraduate project because it demonstrates forms, calculations, data visualisation, responsive design, and client-side persistence without complex infrastructure. This rationale is a project motivation, not a claim that formal user research has already confirmed demand.

## 3. Problem statement

Students and other individuals may find it difficult to keep track of everyday expenses, understand which categories account for most of their recorded spending, and compare expenses against a monthly budget. SpendWise addresses this by providing a simple interface for entering expenses, grouping them into categories, reviewing monthly summaries, and monitoring budget usage.

## 4. Aim and objectives

**Aim:** To design and implement a beginner-friendly, responsive web application for personal expense recording and basic spending analysis.
**Objectives:** (i) build a responsive dashboard; (ii) add and delete expense records; (iii) categorise entries; (iv) calculate monthly totals, counts, averages, and budget remaining; (v) visualise category and daily spending; (vi) provide filters and CSV export; (vii) persist entries in local storage; and (viii) test the core workflows.

## 5. Proposed solution and scope

The dashboard includes summary cards, a daily spending chart, category summaries, a budget progress indicator, and transaction history. Users can enter an expense name, amount, category, and date; search and filter records; change the selected month and budget; delete entries; and export a CSV. The first version is limited to one user in one browser. Bank connections, cloud synchronisation, automated receipt scanning, and multi-user accounts are out of scope.

## 6. Intended users

Potential users include students and individuals who want a lightweight method of recording personal expenses. This audience is proposed rather than validated; interviews or usability testing would be needed to confirm needs and preferences.

## 7. Methodology

The project follows an iterative process: define requirements; design the dashboard layout; implement HTML structure, CSS styling, and JavaScript behaviour; add fictional sample entries; store records in browser local storage; manually test validation, calculations, filtering, deletion, export, responsiveness, and persistence; then deploy the static site if required. No experimental outcomes or user feedback are claimed before testing is conducted.

## 8. Technologies used

**HTML5:** semantic structure and forms. **CSS3:** responsive layout, typography, contrast, and focus states. **Vanilla JavaScript:** validation, calculations, chart drawing, filters, local storage, and CSV export. **Browser local storage:** persistence in the same browser. **GitHub Pages:** optional static hosting.

## 9. Project deliverables

The deliverables are the working website (*index.html*), setup and deployment guide (*README.md*), formal report (*PROJECT_REPORT.md*), manual test plan (*TESTING_CHECKLIST.md*), this combined Lean Canvas and Concept Note PDF, and an editable PowerPoint presentation.

## 10. Expected outcomes

The expected outcome is a static website that lets a user enter expenses, view monthly summaries, compare categories, monitor a selected budget, and export transactions. It should demonstrate core web development skills. These are intended outcomes, not claims of a completed formal evaluation.

## 11. Limitations, ethics, and risks

Entries are manually supplied and may be incomplete. Local storage is limited to the current browser/device and may be cleared; it does not provide accounts, backups, or cross-device sync. Starter entries are fictional. The app does not connect to banks or provide professional financial advice. Users should not enter sensitive financial information. If real user data is collected in a future version, the project should minimise collection, obtain appropriate consent, provide a privacy notice, and apply suitable security and legal safeguards. Budget messaging should remain neutral and non-judgemental.

## 12. Future enhancements

Possible extensions include recurring transactions, savings goals, custom categories, downloadable monthly reports, optional CSV import, and more formal accessibility and usability testing. Secure accounts and synchronisation would require a backend, authentication, access controls, secure data handling, and operating costs. Bank integrations would require an authorised provider, consent, and a security and compliance review. These are future possibilities, not current features.

## 13. Conclusion

SpendWise demonstrates how HTML, CSS, and vanilla JavaScript can be combined to create a responsive expense dashboard. Manual entry, monthly summaries, category analysis, budget tracking, and CSV export provide a practical undergraduate project. The current version remains a local browser prototype with fictional starter data. Further testing and user feedback are needed before conclusions can be drawn about usability or real-world effectiveness.

## Validation status

No customer interviews, survey results, adoption metrics, financial outcomes, or experimental results are claimed in this document. The Lean Canvas records assumptions and possible future opportunities. Update the report after performing and recording actual tests.